Veet Zaveri

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Alexandria, VA

Research Interests

Machine learning, with interests in robustness and generalization, model evaluation, representation and multimodal learning, and foundation models. Research experience spans computer vision, language models, and time-series modeling.

Education

Johns Hopkins University — **Baltimore, MD**
Master of Science in Computer Science — *Expected May 2027*
GPA: 4.00/4.00

DePaul University — **Chicago, IL**
Bachelor of Science in Business Management with Honors — *June 2021*
GPA: 3.99/4.00

Recognition

IDA 3-Star Analysis Award — Sole recipient of the organization's highest-tier analysis award during the six-month award cycle.

Publications

1. **Zaveri, V.**, & Maiya, A. S. (2026). "Towards a Better Characterization of Adversarial Attacks in Geospatial Imagery." *Remote Sensing*, 18(12), 2041. DOI: 10.3390/rs18122041.
2. Goyal, A.[*], **Zaveri, V.**[*], Harris, C., & Stevens, R. D. (2026). "Non-invasive Intracranial Pressure Waveform Reconstruction with Deep Learning." *medRxiv*. Under review at *npj Digital Medicine*. DOI: 10.64898/2026.06.07.26354958.
3. Jardon, C.[†], Sheppard, B.[†], & **Zaveri, V.**[†] (2023). "A Motion Planning Algorithm in a Figure Eight Track." *The PUMP Journal of Undergraduate Research*, 6, 224–249. DOI: 10.46787/pump.v6i0.3559.

[*]Equal contribution and shared first authorship. [†]All authors contributed equally.

Ongoing Research

Non-invasive ICP Waveform Reconstruction Using ECG and PPG ***Manuscript in preparation***
Investigating ECG-only, PPG-only, and ECG+PPG models for non-invasive ICP reconstruction, with early- and late-fusion strategies and waveform-aware objectives incorporating reconstruction, correlation, and derivative information.

Domain-Aware Hyperparameter Selection Under Distribution Shift ***Ongoing research***
Evaluating how standard versus domain-aware validation objectives affect hyperparameter selection and performance on unseen domains; comparing particle swarm optimization and random search on PACS and Multi-Domain Sentiment benchmarks. *Student abstract submitted to the AAAI Student Abstract and Poster Program; decision pending.*

Research Experience

Institute for Defense Analyses (IDA) — **Alexandria, VA**
Research Associate (CS Researcher / Data Scientist) — *Jan. 2024 – Present*

- **Led and first-authored** a systematic study of **20+ CNN and Transformer architectures** for detecting adversarial and generative manipulations of satellite imagery under domain shift, previously unseen attacks, and low-prevalence screening; **published in *Remote Sensing***.
- Designed controlled evaluations of conventional, Earth-observation-pretrained, and vision-language models; analyzed cross-domain transfer, false-alarm burden, and performance under limited target-domain data.

- Developed a machine-learning-powered, parallelized Python acquisition and cleaning pipeline that processed **21M+ records**, increased ingestion throughput by **50×**, and supported **50+ analytical products for senior U.S. government sponsors**; selected as the **sole recipient of IDA's highest-tier 3-Star Analysis Award** for the six-month award cycle.
- Evaluated frontier LLMs through **AWS Bedrock** for structured information extraction and data cleaning; achieved **90%+ extraction accuracy** and a hallucination rate below **5%** on a 30+ document benchmark, and applied LLMs to classify **10K+ records**.
- Built and deployed an automated data pipeline and **Streamlit application** covering **26K+ records** across multiple sources, with automated weekly refreshes and adoption by government sponsors and **20+ internal users**.
- Built a semantic document-analysis system over **500+ documents** using sentence-transformer embeddings and cosine similarity to identify semantically overlapping content, reducing manual analysis by **75+ hours**.
- Built an NLP application that classifies **300+ article titles daily** and generates summaries using ktrain and txtai, reducing manual literature-monitoring workloads by **20+ hours per week**.
- Clustered and linked **2,000+ entity names** across heterogeneous datasets using TF-IDF n-grams, sentence-transformer embeddings, and topic modeling for cross-source entity resolution and downstream analysis.
- Engineered and optimized **10+ TB of SQL data infrastructure** on Linux across 6+ data sources using indexing, partitioning, and database tuning to eliminate API bottlenecks; increased data retrieval throughput by **20×**, from approximately 10K to 200K records per day.

LCICM – Johns Hopkins University **Remote**

Graduate Deep Learning Researcher *Aug. 2024 – Present*

- **Co-first-authored** a study of non-invasive intracranial-pressure (ICP) waveform reconstruction using **158 adult patients** and approximately **5,322 hours of waveform data**, contributing to methodology, implementation, analysis, and manuscript preparation.
- Developed and compared **LSTM, GRU, multiscale V-Net, TCN, and Transformer-based models** for ICP reconstruction using arterial blood pressure, photoplethysmography, and ECG; evaluated early/late fusion and waveform-aware training losses.
- Conducted, to our knowledge, the **first and largest external validation to date** of ML-based noninvasive ICP waveform reconstruction, using a **47-patient internal cohort** and an independent **111-patient external cohort**; across five architectures, external performance ranged from **4.276–4.946 mmHg MAE** and **0.599–0.722 Pearson correlation**.
- Built the end-to-end physiological signal-processing pipeline used to synchronize concurrent intracranial and extracranial waveforms, construct **30-second prediction windows** with historical context, and resample signals to **125 Hz** for model-ready tensors.
- Developed automated signal-quality filtering for flatline, noisy, periodic, and physiologically implausible segments using amplitude, variability, heart-rate plausibility, and signal-quality checks.

Wilbur Wright College **Chicago, IL**

Topology Student Researcher *Jan. 2022 – Jun. 2023*

- Proved the two-robot figure-eight motion-planning problem has **topological complexity 3** and constructed a collision-free algorithm using three continuous motion-planning instructions.
- Used configuration-space topology, deformation retraction, and homotopy arguments to characterize motion-planning constraints; **co-authored a peer-reviewed publication in *The PUMP Journal of Undergraduate Research*.**

Research Leadership and Mentoring

LCICM – Johns Hopkins University **Remote**

Lab Director *Sep. 2026 – Present*

- Help supervise **5 research projects involving 11 students**, providing guidance on research planning, technical challenges, execution, and next-step priorities.
- Coordinate weekly laboratory meetings and project reviews; support interdisciplinary research operations and IRB-related administration under PI Robert D. Stevens.

Selected Engineering Experience

Mayo Clinic **Remote**

IT Software Engineer Intern *Jul. 2023 – Dec. 2023*

- Contributed to development and internal deployment of a specimen-tracking application using TypeScript, React, HTML, and CSS within an Agile Scrum team.
- Evaluated Next.js and Vite and prepared technical comparisons to guide architecture and technology selection for future software projects.
- Integrated Microsoft Graph API functionality to retrieve and display authenticated user profile information and photos.

Astoria.ai **Remote**

Software Engineer Intern / Architecture Team Lead *Sep. 2022 – May 2023*

- Led the Software Architecture team in evaluating Redis, Kafka, Fluvio, and Hadoop for a talent-management platform aggregating, correlating, and transforming 10M+ records.
- Developed full-stack features with Python, Java, JavaScript, HTML, and CSS across backend data processing and user-facing components.
- Built RESTful ingestion and web-scraping services that collected and processed 100K+ job postings for search, analytics, and candidate-matching workflows.
- Applied regression, clustering, predictive modeling, and neural networks to candidate-matching pipelines connecting recruiters with best-fit candidates.

Teaching Experience

Johns Hopkins University **Baltimore, MD**

Data Structures TA / Grader *Jan. 2023 – May 2023*

- Evaluated Java and Python assignments, laboratories, homework, and quizzes for **20+ students**, assessing correctness, algorithmic reasoning, and data-structure implementation.
- Provided individualized feedback and debugging guidance on **data structures, algorithms, Java, and Python**.

Technical Skills

Programming: Python, Java, TypeScript/JavaScript, SQL

Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, Sentence Transformers, ktrain, txtai

Research Methods: Deep learning, model evaluation and benchmarking, adversarial robustness, distribution-shift analysis, representation learning, multimodal fusion, time-series modeling, computer vision, NLP and language models, foundation-model evaluation

Scientific Computing & Tools: NumPy, Pandas, SciPy, SageMath, LaTeX

Data & Systems: PostgreSQL, SQL Server, MongoDB, ksqlDB, AWS, Git, Docker, Kafka, Redis, Linux